

ARCHITECTURE & PLATFORM UPDATE

The thesis is now a running platform: what it's built from, what shipped this week, and what's next.

An architecture and progress review of StackGraph's implementation — the ingestion substrate, the graph and evidence model, the intelligence layer — and a look at the 109 commits shipped between August 19 and August 21, 2026.

ARCHITECTURE

IMPLEMENTATION

LATEST UPDATES

ROADMAP

From product thesis to running system, in one sprint

The full StackGraph platform — acquisition, normalization, intelligence, and experience — now exists in code, not only in the product brief. The entire repository history spans three days: 109 commits, August 19 through August 21, 2026, building the ingestion substrate, the Postgres + Apache AGE graph, the modernization intelligence layer, and the five product surfaces end to end.

109

Commits, in a 3-day build sprint (Aug 19–21)

39 / 49

API path patterns / operations, contract-frozen

4 of 6

Milestones fully complete; Milestone 4 in progress

5

Product surfaces shipped: Estate, Applications, Technology, Viability, Ask

WHAT THIS UPDATE COVERS

Architecture

The five graph domains, the ingestion pipeline, the Postgres-authoritative + AGE-projection data model, and the monorepo that implements them.

Status

Where the six-milestone build plan actually stands today, lane by lane, against what was scoped.

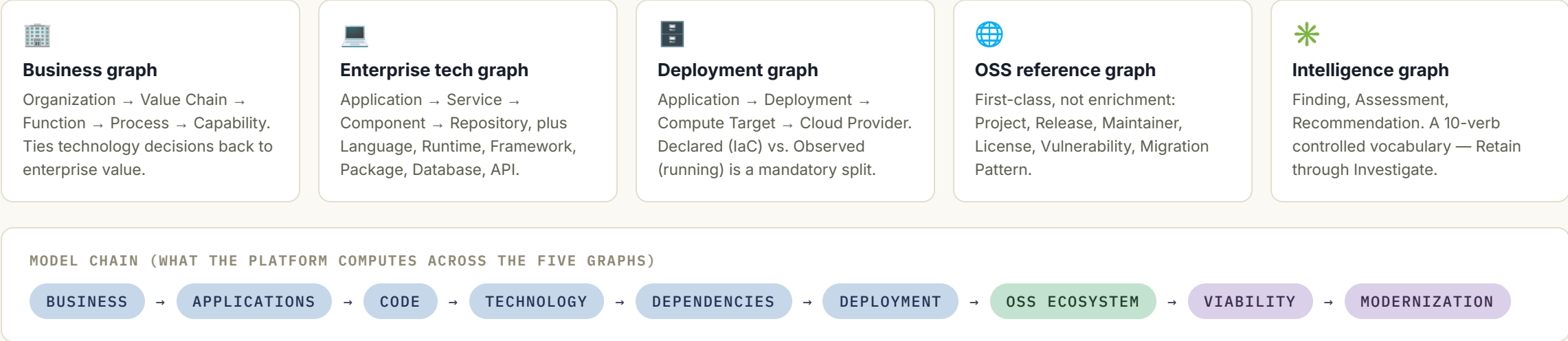
What's next

The concrete gaps between "repository-side complete" and a real customer pilot, plus the V1 intelligence work queued behind it.

"Business → Applications → Code → Technology → Dependencies → Deployment → Infrastructure → OSS ecosystem → Viability → Modernization actions." **This is the model chain the platform now computes end to end** — not a diagram, a running pipeline.

Five graphs, one estate model

Git repositories are a primary sensor, not the final product. Everything StackGraph observes is normalized into five interconnected graph domains, bridged by shared concepts — Package, Technology, Runtime, Framework, Capability — so a dependency finding can be traced all the way up to a business process.



Four coordinated systems, three scanner passes

Acquisition, normalization, intelligence, and experience run as separate systems with independent failure and release boundaries. Every connected repository moves through three scanner passes that widen from fast inventory to asynchronous depth — the 20-minute pilot objective is time to first useful evidence, not time to finish Pass C.

PASS	WHAT IT DOES	EVIDENCE PRODUCED
Pass A — Fast inventory	Manifests, lockfiles, Dockerfiles, K8s and Terraform files across 100+ repos quickly	Declared technologies, dependencies, deployment intent
Pass B — Evidence refinement	AST / import usage analysis; declared-vs-referenced comparison	Observed usage, unused-dependency candidates
Pass C — Async depth	Transitive neighborhoods, capability inference, internal reuse, migration history	Inferred capability mapping, reuse and migration signal

SOURCES WIRED TODAY

GitHub App

Webhook-driven (push, installation, repository events) + periodic reconciliation, contents/metadata read-only.

npm registry + deps.dev v3

Resolves effective registry origin from lockfile + .npmrc; deps.dev supplies resolved graphs, licenses, deprecations.

OSV

Batch vulnerability queries against exact observed package-version purls; canonical ID with CVE/GHSA aliases.

Curated + external seed

192-entry curated technology taxonomy, plus a 5,232-package Hugging Face npm OSS catalog import. PyPI parity is pending.

Postgres is the record; the graph is a projection

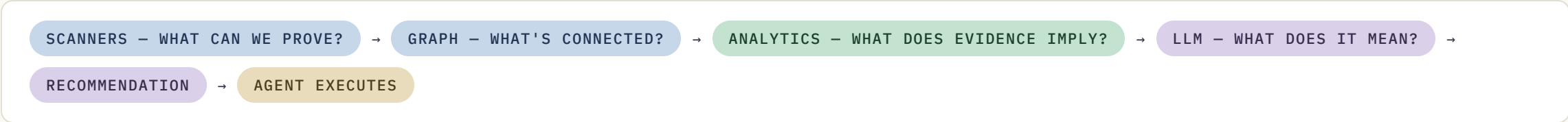
PostgreSQL is the authoritative system of record. Apache AGE is an asynchronous graph-traversal projection, not the source of truth: projection workers claim outbox rows with a lease and commit AGE writes in the same transaction as acknowledgement. Read paths fall back to tenant-filtered SQL automatically if the graph projection is stale.

EVERY FACT, RELATIONSHIP, AND RECOMMENDATION CARRIES AN EVIDENCE TIER

DECLARED Stated in a manifest, lockfile, or IaC file	OBSERVED Confirmed by AST import or runtime signal	INFERRED Derived from naming, routes, or combined signals	CURATED A versioned, provenanced human knowledge seed	EXTERNAL Measured from deps.dev, OSV, or public OSS data
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"No material conclusion without evidence." Sub-0.85 confidence joins render as a dashed **possible match** with Confirm / Reject controls — never as a settled fact. Scanners establish what can be proven; the LLM layer never manufactures estate facts.

REASONING PIPELINE (AN EXPLICIT GUARDRAIL AGAINST BECOMING AN OPAQUE LLM HEURISTIC SYSTEM)



One monorepo, four service domains

A pnpm-workspace monorepo separates the API and UI apps from the discovery, data-platform, and intelligence services, with shared contracts and design tokens as first-class packages.

REPOSITORY LAYOUT

```
apps/  
  api/           FastAPI backend – routes, auth, AGE graph, AI ask, read models  
  web/           Next.js UI – estate, applications, business-map, ask, admin  
services/  
  data-platform/  deps.dev + OSV workers, AGE projector, catalog, seed  
  enterprise-discovery/  GitHub App, webhooks, repository scanner, evidence store  
  intelligence/  
    ai-services/  capabilities, modernization, provider-neutral AI layer  
packages/  
  design-system/  Strata tokens and primitives  
  graph-ui/       Bounded, read-only graph canvas (React Flow + Dagre)  
  shared/         Generated OpenAPI types, API client, fixtures  
stackgraph-foundation/  schema.sql, ontology, contracts v1  
infrastructure/   DB migrations, observability, production Caddy/TLS
```

TECH STACK

Frontend	Next.js / TypeScript, App Router
API	FastAPI (Python), generated OpenAPI
Database	PostgreSQL 16 — authoritative store
Graph	Apache AGE 1.6 — async projection
Queue	Postgres-backed leases, no separate broker
AI layer	Provider-neutral: OpenRouter, OpenAI, Anthropic
Auth	OIDC (PKCE), rotating sessions, RBAC ladder
Deploy	Docker Compose, Caddy TLS, MinIO/S3
Observability	Prometheus, Alertmanager, Grafana
Testing	Playwright e2e, WCAG 2.2 AA gate

Milestone 4 of 6: pilot readiness

Contract hardening, the ingestion substrate, the first vertical estate slice, and the V0 product experience are complete. Pilot readiness is implemented end to end in code — what remains is a real installation and human sign-off, not more building.

M	MILESTONE	STATUS
0	Contract hardening	Complete — canonical OpenAPI (39 paths / 49 ops), generated TS types, drift check, 20 contract fixtures
1	Running ingestion substrate	Complete — continuous GitHub / deps.dev / OSV / projection / intelligence workers
2	First vertical estate slice	Complete in code — GitHub App token broker, signed webhooks, Docker/K8s/Terraform facts; hosted callback is a deployment task
3	V0 product experience	Complete — 5 surfaces, reviews/health/admin/evidence/graph lenses, cross-browser WCAG 2.2 AA
4	Pilot readiness	Repository-side complete; target acceptance pending — a real 100+ repo pilot has not yet run
5	V1 intelligence	Not started — business-capability mapping, entropy, portfolio ranking

LANE A

Data platform / OSS
Schema, evidence, deps.dev, OSV, AGE projection tested.
Remaining: production deploy, object-storage lifecycle, PyPI.

LANE B

Enterprise discovery
GitHub install/reconciliation, scanning, registry resolution tested. Remaining: hosted callback, pilot-scale proof.

LANE D

Intelligence / quality
Taxonomy, inference, duplicate detection, AI Ask, RLS tested.
Remaining: tenant policy rollout, calibration targets.

What shipped, August 19 – 21

The entire repository history is 109 commits inside a 72-hour build window. Grouped by theme, most-recent first:

DATE	THEME	SHIPPED
Aug 21 (today)	Estate & technology unification	Application dependency hierarchy, grouped Technology Explorer, unified estate exploration and intelligence views
Aug 20	Admin & Docker operability	One-command Docker startup, Admin service start/stop controls, GitHub repository picker, worker heartbeat visibility
Aug 20	Production hardening	Largest single commit (58 files): OIDC auth, TLS via Caddy, object storage, Prometheus/Alertmanager/Grafana, release & security CI gates
Aug 20	Pilot readiness gates	Scripted synthetic 100-repository pilot run, Playwright primary-journey e2e test, pilot runbook
Aug 19–20	Business Map persistence	Converted from browser-localStorage-only to server-backed CRUD with RBAC gating and tenant isolation — closed the top Lane C risk
Aug 19	Foundation & intelligence	Backend data foundation, AGE projection, GitHub acquisition, deps.dev/OSV/npm lanes, capability & modernization intelligence (Phases 1–3)

Repository-side work is done; external validation is next

The production-readiness review found six deployment blockers — no auth entry point, no web container image, no deployment topology as code, no delivery pipeline, no observability, no rate limiting. All six have since been closed. What's left is validation, not construction.

BEFORE PRODUCTION ACCEPTANCE

- Real IdP / GitHub App provisioning with a push-to-live-UI trace
- Cloud object-store durability — S3 versioning, Object Lock, KMS (local MinIO only today)
- A representative 100+ repository pilot run with an archived report — only a synthetic run has passed so far
- Human and security sign-off: keyboard/screen-reader sessions, penetration test, privacy and retention approval
- Resolve one legacy local database migration-checksum quirk

NEXT: V1 INTELLIGENCE (MILESTONE 5)

- ◆ Business-capability mapping, beyond today's UI-first pass
- ◆ Technology entropy scoring, by domain
- ◆ Internal reuse detection and consolidation ranking
- ◆ Modernization portfolio ranking across the estate
- ◆ Calibration corpus and outcome feedback loop
- ◆ Bounded ecosystem expansion — PyPI parity first, then Maven / Cargo / NuGet

The pilot's north star hasn't changed: **connect a GitHub organization of 100+ repositories and, within roughly twenty minutes, surface at least five material facts or opportunities engineering leadership didn't already know.** That test is now scripted and gate-checked — synthetic runs pass at 100% evidence coverage. A real customer pilot has not yet happened.

What compounds beyond the scanner

Repository scanning is necessary infrastructure and reproducible by a competitor in a quarter. What compounds over years — and is now genuinely accumulating, not just planned — is the evidence behind these six assets.

ENT

Enterprise software graph
A continuously refreshed model tying the org's software to business value — now server-backed end to end.

OSS

OSS reference graph
A normalized external model of technology purpose, health, and trajectory — seeded with 5,232 curated packages.

OSS

Migration graph
Observed evidence of how real projects migrate between technologies — flagged as possibly the strongest proprietary asset.

INT

Capability graph
A semantic model that reasons about function, not package identity — "function before package."

ENT

Internal reuse graph
Knowledge of proven capability implementations already inside the org, surfaced before a rebuild is suggested.

INT

Outcome feedback loop
Recommendation → accepted/rejected → migration → rescan
→ observed outcome, sharpening quality over time.

Software creation is becoming dramatically easier. Understanding the software estate is not. Three days in, the platform that does that understanding now runs — this deck is the record of what's built, and what's honestly still left.